

Solution Key: Human Capital Investment Decision

Given Information

Johny is considering a **3-year college diploma**.

- Length of program: 3 years
 - Annual tuition: \$1,200
 - Annual books/materials: \$300
 - Earnings without college: \$8,000 per year
 - Earnings with diploma: \$14,000 per year
 - Working life after graduation: 12 years
 - Market interest rate: $r = 6\% = 0.06$
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1. Total annual direct cost of attending college

The annual **direct cost** is:

$$\text{Direct Cost} = \text{Tuition} + \text{Books}$$

$$\text{Direct Cost} = 1200 + 300 = 1500$$

Annual direct cost = \$1,500

2. Opportunity cost per year of attending college

The **opportunity cost** is the income Johny gives up by going to school instead of working.

$$\text{Opportunity Cost per year} = \$8,000$$

Opportunity cost per year = \$8,000

3. Total cost of the investment over the 3-year period (ignoring discounting)

The total annual cost is:

$$\text{Annual Total Cost} = \text{Direct Cost} + \text{Opportunity Cost}$$

$$\text{Annual Total Cost} = 1500 + 8000 = 9500$$

Since the program lasts 3 years:

$$\text{Total Cost over 3 years} = 3 \times 9500 = 28,500$$

Total undiscounted cost over 3 years = \$28,500

4. Annual earnings gain from completing the diploma

The additional annual earnings from the diploma are:

$$\Delta Y = 14,000 - 8,000 = 6,000$$

Annual earnings gain = \$6,000

5. Present value of benefits

Johnny receives the additional earnings of \$6,000 each year for 12 years after graduation.

So the present value of benefits, measured **at the time of graduation**, is the present value of a 12-year annuity:

$$PV_B^{(t=3)} = 6000 \left(\frac{1 - (1.06)^{-12}}{0.06} \right)$$

First compute:

$$(1.06)^{12} \approx 2.0122$$

$$(1.06)^{-12} \approx \frac{1}{2.0122} \approx 0.49697$$

Then:

$$\frac{1 - 0.49697}{0.06} = \frac{0.50303}{0.06} \approx 8.3838$$

So:

$$PV_B^{(t=3)} = 6000(8.3838) \approx 50,302.8$$

This is the present value **at the time Johnny finishes school**. Since the schooling decision is made **today**, we must discount this amount back 3 years:

$$PV_B^{(t=0)} = \frac{50,302.8}{(1.06)^3}$$

$$(1.06)^3 = 1.191016$$

$$PV_B^{(t=0)} \approx \frac{50,302.8}{1.191016} \approx 42,235.9$$

$PV(\text{Benefits}) \approx \$42,236$

6. Present value of costs

Johnny incurs total annual cost of \$9,500 for 3 years. These costs occur during years 1, 2, and 3.

Thus the present value of costs is:

$$PV_C = \frac{9500}{1.06} + \frac{9500}{(1.06)^2} + \frac{9500}{(1.06)^3}$$

Compute each term:

$$\frac{9500}{1.06} \approx 8962.26$$

$$\frac{9500}{1.1236} \approx 8454.97$$

$$\frac{9500}{1.191016} \approx 7974.71$$

Add them:

$$PV_C \approx 8962.26 + 8454.97 + 7974.71 = 25,391.94$$

$$PV(\text{Costs}) \approx \$25,392$$

7. Net Present Value (NPV) decision

The net present value is:

$$NPV = PV(\text{Benefits}) - PV(\text{Costs})$$

$$NPV = 42,235.9 - 25,391.94 = 16,843.96$$

$$NPV \approx \$16,844$$

Since:

$$NPV > 0$$

Johny **should invest** in the diploma.

$$\boxed{\text{Yes, Johny should undertake the investment.}}$$

8. Effect of a higher interest rate ($r = 8\%$)

If the market interest rate rises, future earnings are discounted more heavily.

This means:

- The present value of future benefits falls.
- The attractiveness of education decreases.

Now compute the present value of benefits at 8%:

$$PV_B^{(t=3)} = 6000 \left(\frac{1 - (1.08)^{-12}}{0.08} \right)$$

$$(1.08)^{12} \approx 2.5182$$

$$(1.08)^{-12} \approx 0.3971$$

$$\frac{1 - 0.3971}{0.08} = \frac{0.6029}{0.08} \approx 7.5363$$

$$PV_B^{(t=3)} \approx 6000(7.5363) = 45,217.8$$

Discounting back 3 years:

$$PV_B^{(t=0)} = \frac{45,217.8}{(1.08)^3}$$

$$(1.08)^3 = 1.2597$$

$$PV_B^{(t=0)} \approx \frac{45,217.8}{1.2597} \approx 35,895.7$$

Now compute the present value of costs:

$$PV_C = \frac{9500}{1.08} + \frac{9500}{(1.08)^2} + \frac{9500}{(1.08)^3}$$

$$\frac{9500}{1.08} \approx 8796.30$$

$$\frac{9500}{1.1664} \approx 8144.15$$

$$\frac{9500}{1.2597} \approx 7541.48$$

$$PV_C \approx 8796.30 + 8144.15 + 7541.48 = 24,481.93$$

Thus:

$$NPV_{8\%} \approx 35,895.7 - 24,481.93 = 11,413.77$$

So the NPV is still positive, but smaller.

A higher interest rate reduces the attractiveness of education, but Johnny should still invest.

9. Approximate internal rate of return (IRR)

The IRR is the discount rate that makes:

$$PV(\text{Benefits}) = PV(\text{Costs})$$

We already know:

- At 6%, NPV is positive.
- At 8%, NPV is still positive.

So the IRR must be **above 8%**.

Try 12%:

$$PV_B^{(t=3)} = 6000 \left(\frac{1 - (1.12)^{-12}}{0.12} \right)$$

$$(1.12)^{12} \approx 3.89598$$

$$(1.12)^{-12} \approx 0.2567$$

$$\frac{1 - 0.2567}{0.12} = \frac{0.7433}{0.12} \approx 6.194$$

$$PV_B^{(t=3)} \approx 6000(6.194) = 37,164$$

Discount back 3 years:

$$PV_B^{(t=0)} = \frac{37,164}{(1.12)^3}$$

$$(1.12)^3 = 1.404928$$

$$PV_B^{(t=0)} \approx 26,453$$

Now costs at 12%:

$$PV_C = \frac{9500}{1.12} + \frac{9500}{(1.12)^2} + \frac{9500}{(1.12)^3}$$

$$\approx 8482.14 + 7573.34 + 6761.91 = 22,817.39$$

So:

$$NPV_{12\%} \approx 26,453 - 22,817 = 3,636$$

Still positive, so IRR is above 12%.

Try 15%:

Benefits:

$$PV_B^{(t=3)} = 6000 \left(\frac{1 - (1.15)^{-12}}{0.15} \right)$$

$$(1.15)^{12} \approx 5.3503, \quad (1.15)^{-12} \approx 0.1869$$

$$\frac{1 - 0.1869}{0.15} \approx 5.4207$$

$$PV_B^{(t=3)} \approx 6000(5.4207) = 32,524.2$$

Discount back 3 years:

$$(1.15)^3 = 1.520875$$

$$PV_B^{(t=0)} \approx \frac{32,524.2}{1.520875} \approx 21,385$$

Costs:

$$PV_C = \frac{9500}{1.15} + \frac{9500}{(1.15)^2} + \frac{9500}{(1.15)^3}$$

$$\approx 8260.87 + 7183.37 + 6246.41 = 21,690.65$$

Now NPV is slightly negative:

$$NPV_{15\%} \approx 21,385 - 21,691 \approx -306$$

So the IRR is between 14% and 15%, very close to 15%.

$$IRR \approx 14.8\% \text{ to } 15\%$$

10. Comparative statics

(a) If expected post-education earnings fall

Then:

$$\Delta Y = \text{earnings with diploma} - \text{earnings without diploma}$$

becomes smaller.

Therefore:

- present value of benefits falls,
- NPV falls,
- IRR falls.

So Johnny becomes **less likely** to invest.

Lower post-education earnings reduce the attractiveness of schooling.

(b) If tuition fees increase

Higher tuition raises direct cost D , so:

- marginal cost rises,
- present value of costs rises,
- NPV falls.

Thus Johnny becomes **less likely** to invest.

Higher tuition reduces the incentive to invest in education.

(c) If his working life becomes shorter

A shorter working life means fewer years over which to earn the higher post-education income.

Thus:

- present value of benefits falls,
- NPV falls,
- IRR falls.

So Johnny becomes **less likely** to invest.

A shorter working life reduces the return to education.

Final Conclusion

Johnny should invest in the 3-year diploma because:

$$PV(\text{Benefits}) > PV(\text{Costs})$$

and

$$NPV > 0$$

The estimated internal rate of return is approximately:

$IRR \approx 15\%$

which is well above the market interest rate of 6%.

Therefore, the diploma is a worthwhile human capital investment.